



LLM Training and NVIDIA NeMo AutoModel

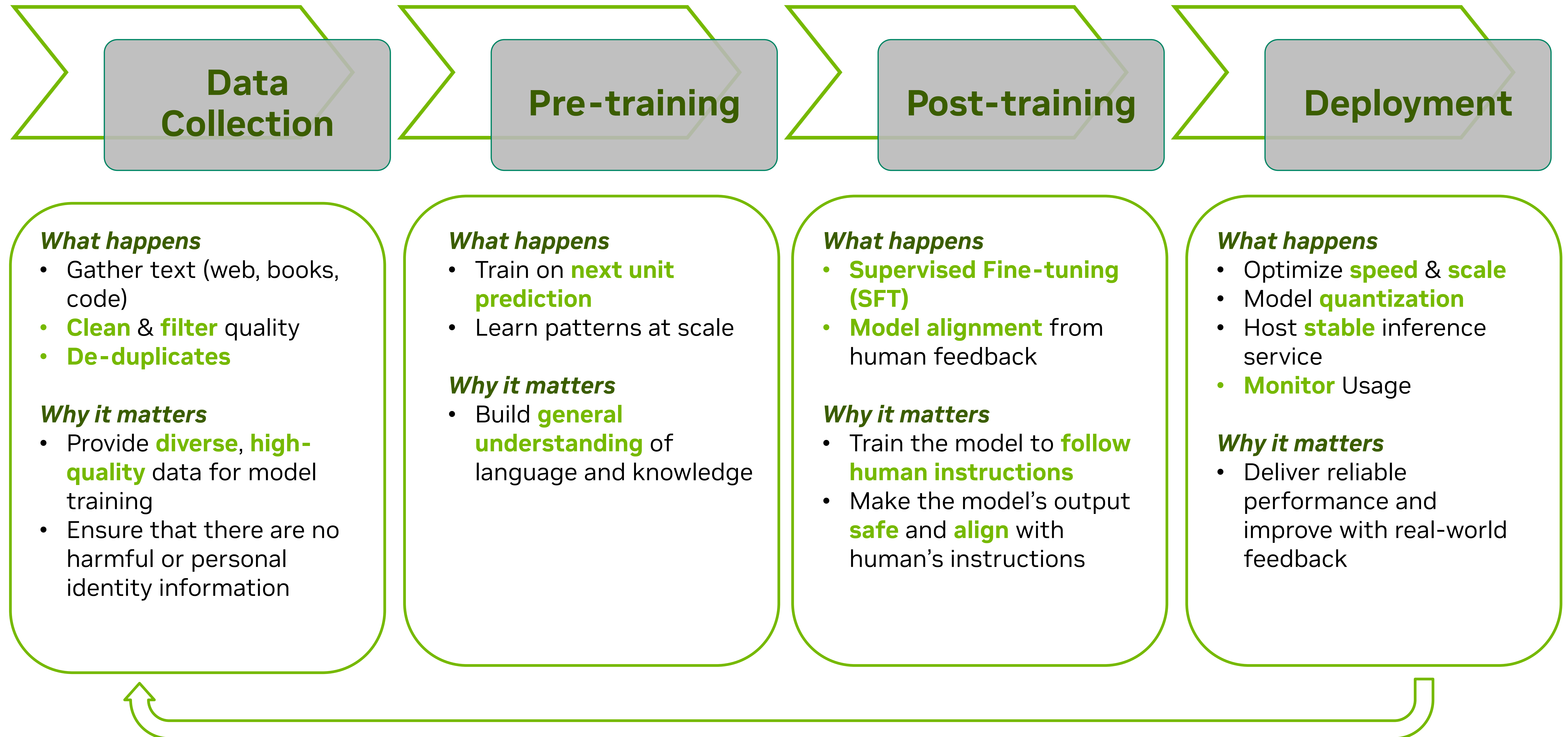
A Beginner's Guide to Pre-training and SFT

Iven Fu | Solutions Architect, NVAITC



LLM Training Phases

LLMs Life Cycle



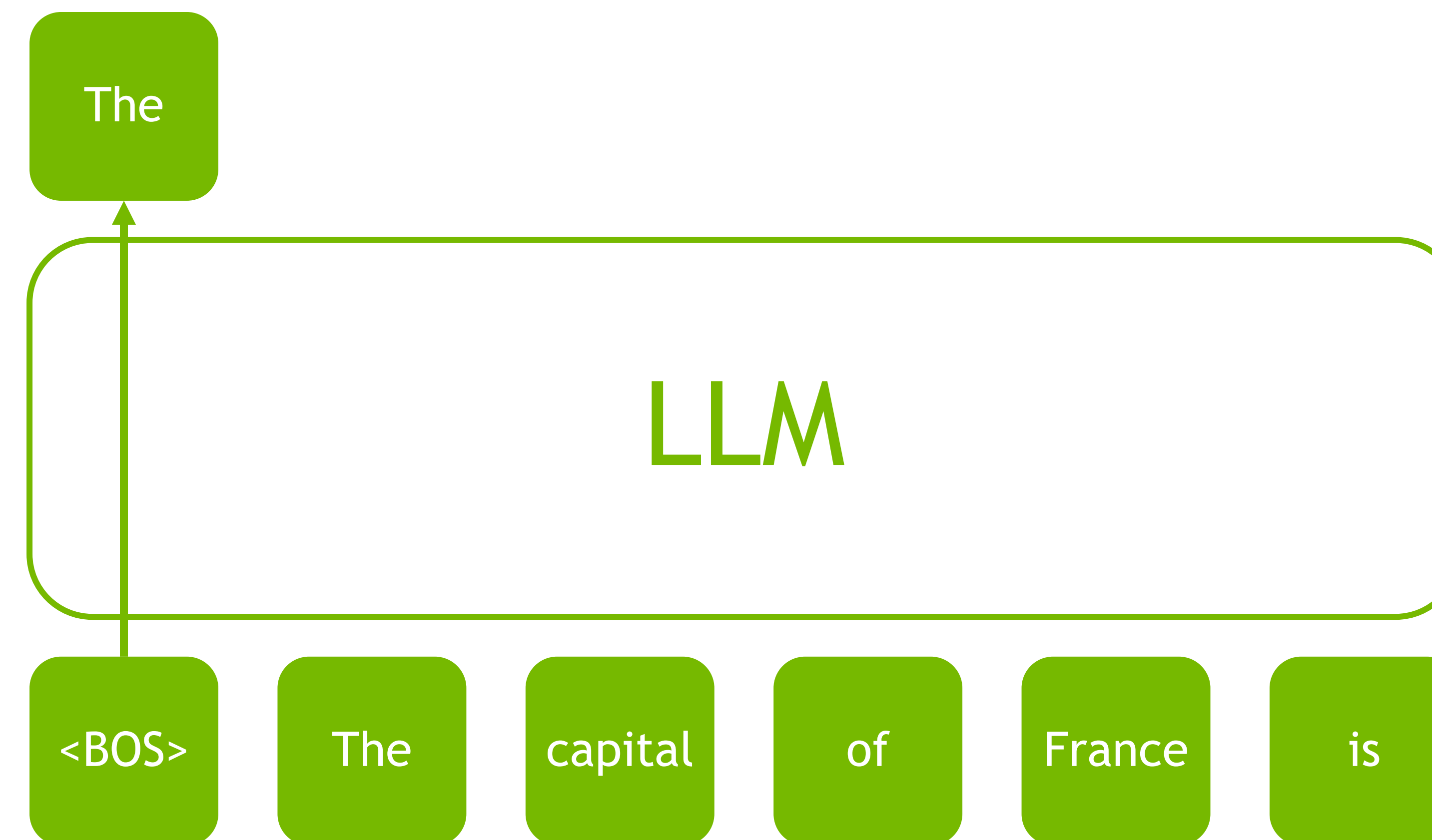
Pre-training

- Learning from **large-scale unlabeled data** (about several **trillion** tokens)
- **Self-supervised** learning through **Next Unit Prediction**
- Learns patterns, language, and **world knowledge**
 - **Grammar & syntax**
 - Facts about the world



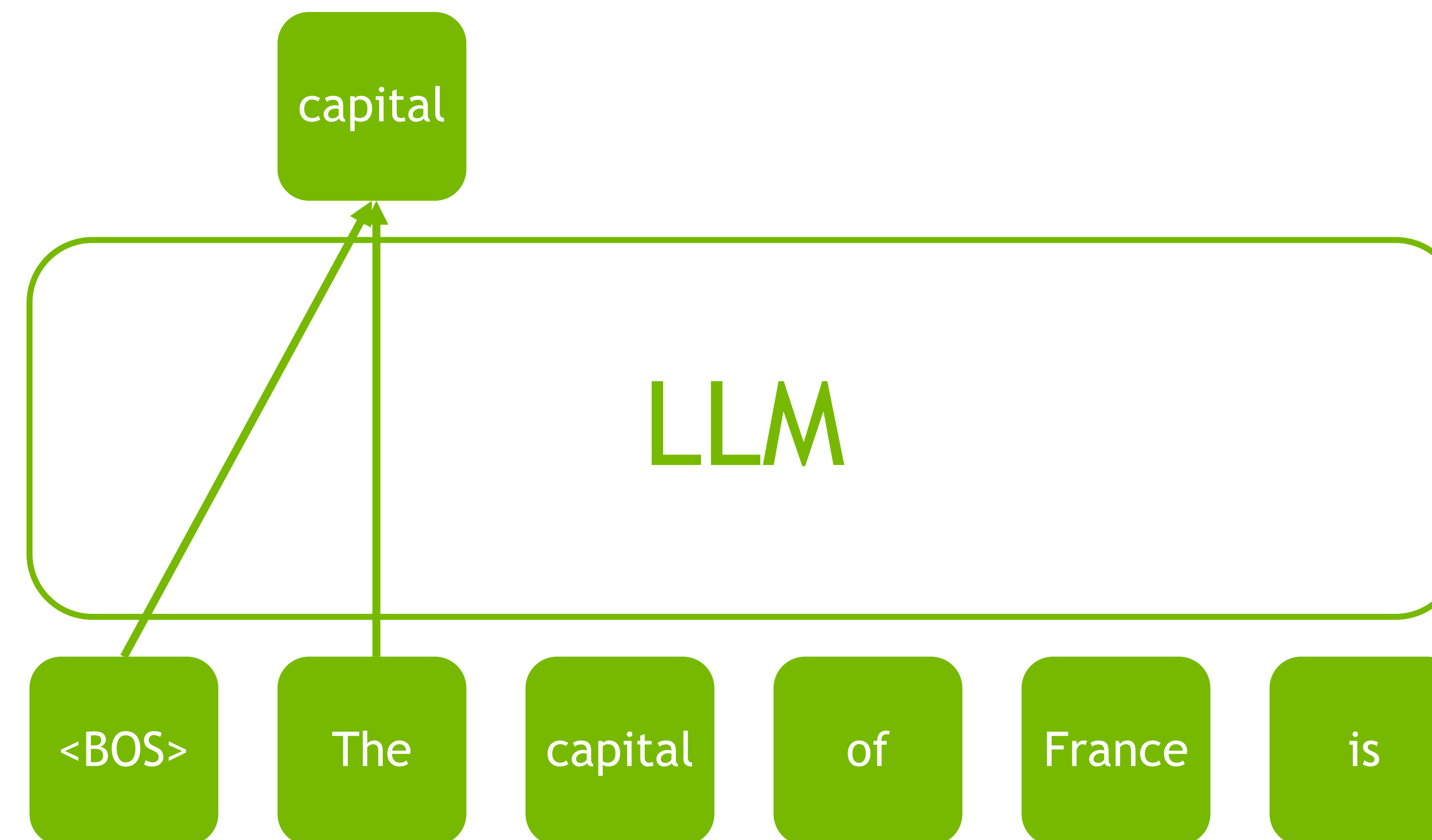
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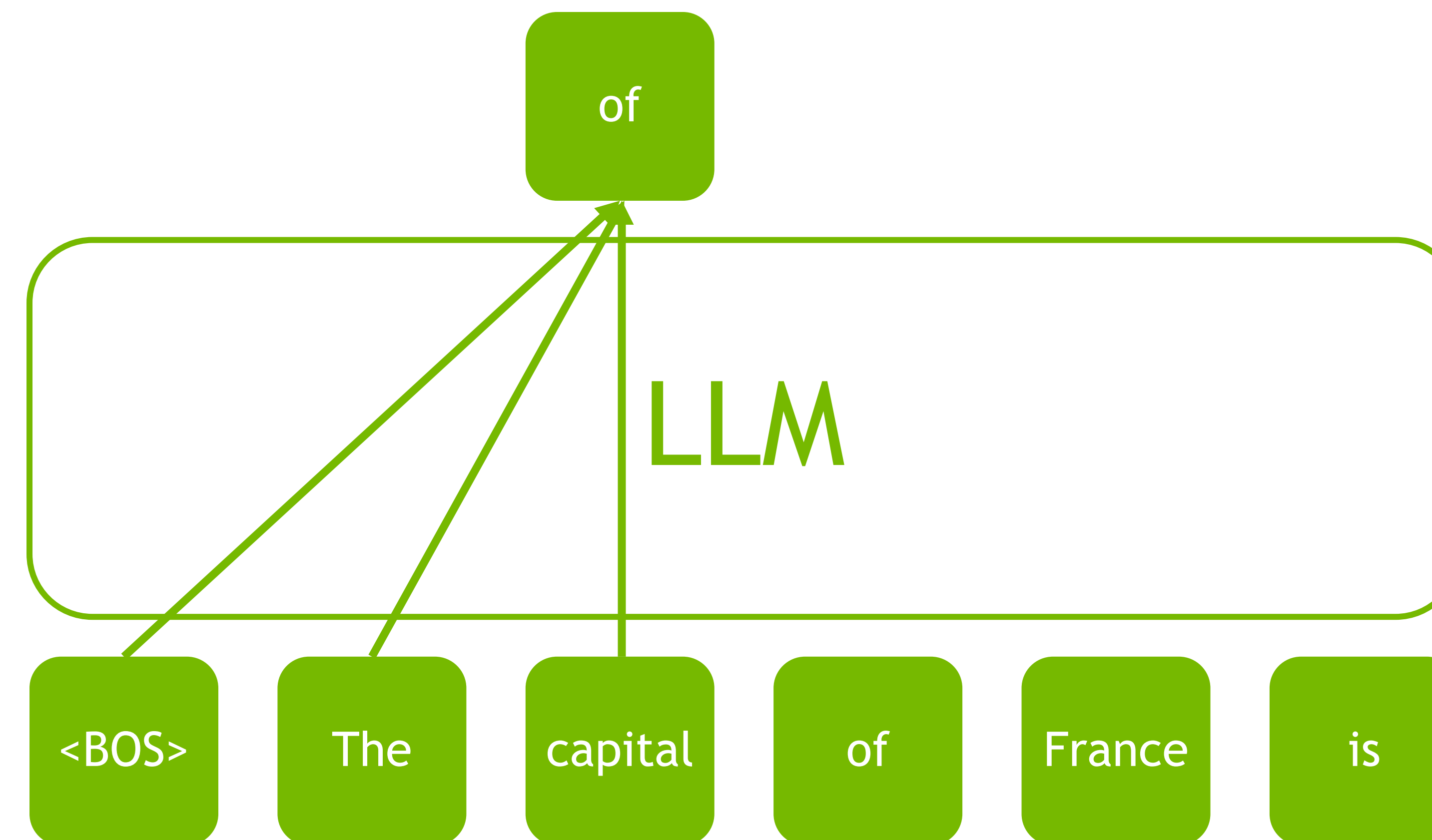
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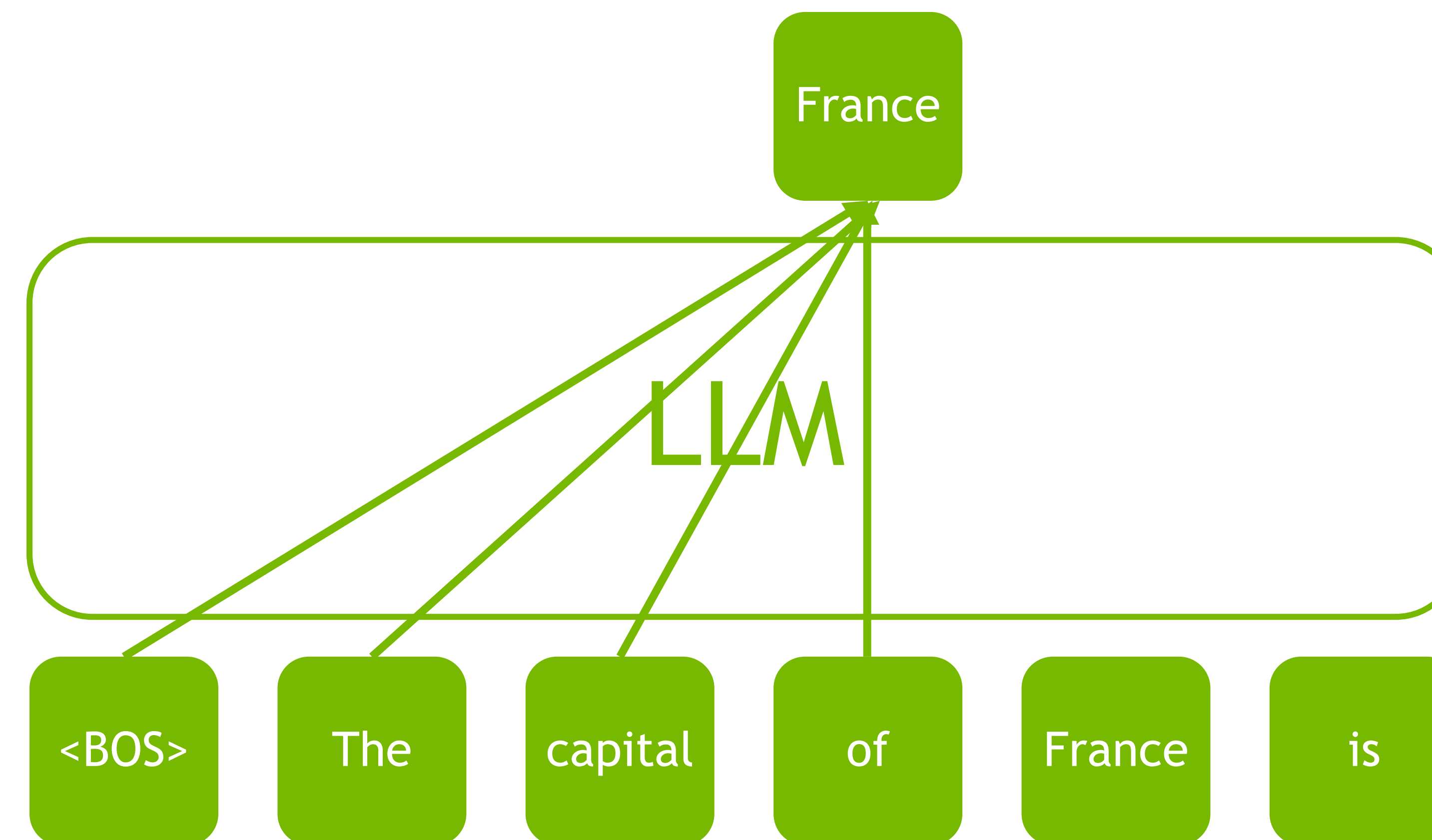
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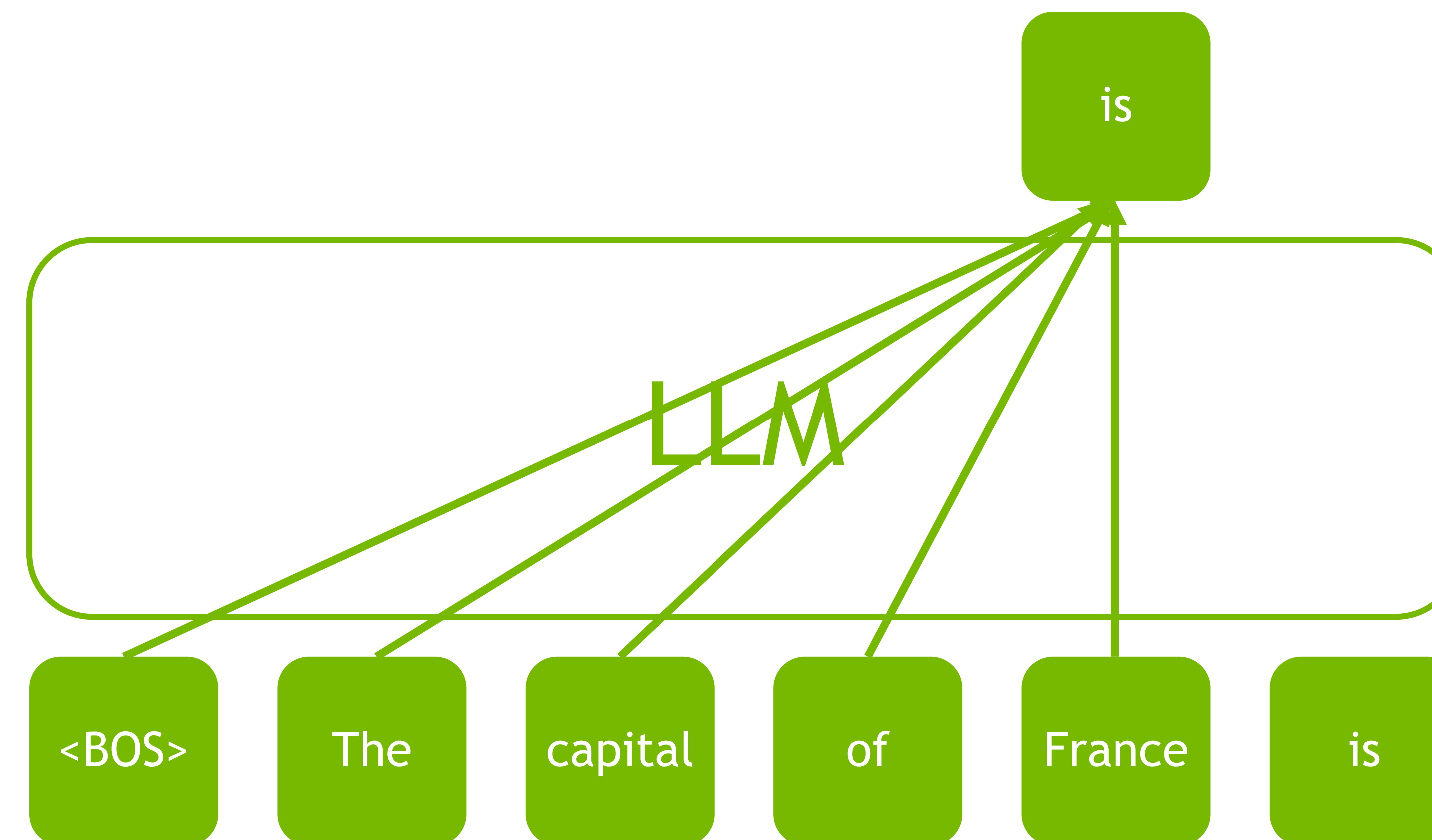
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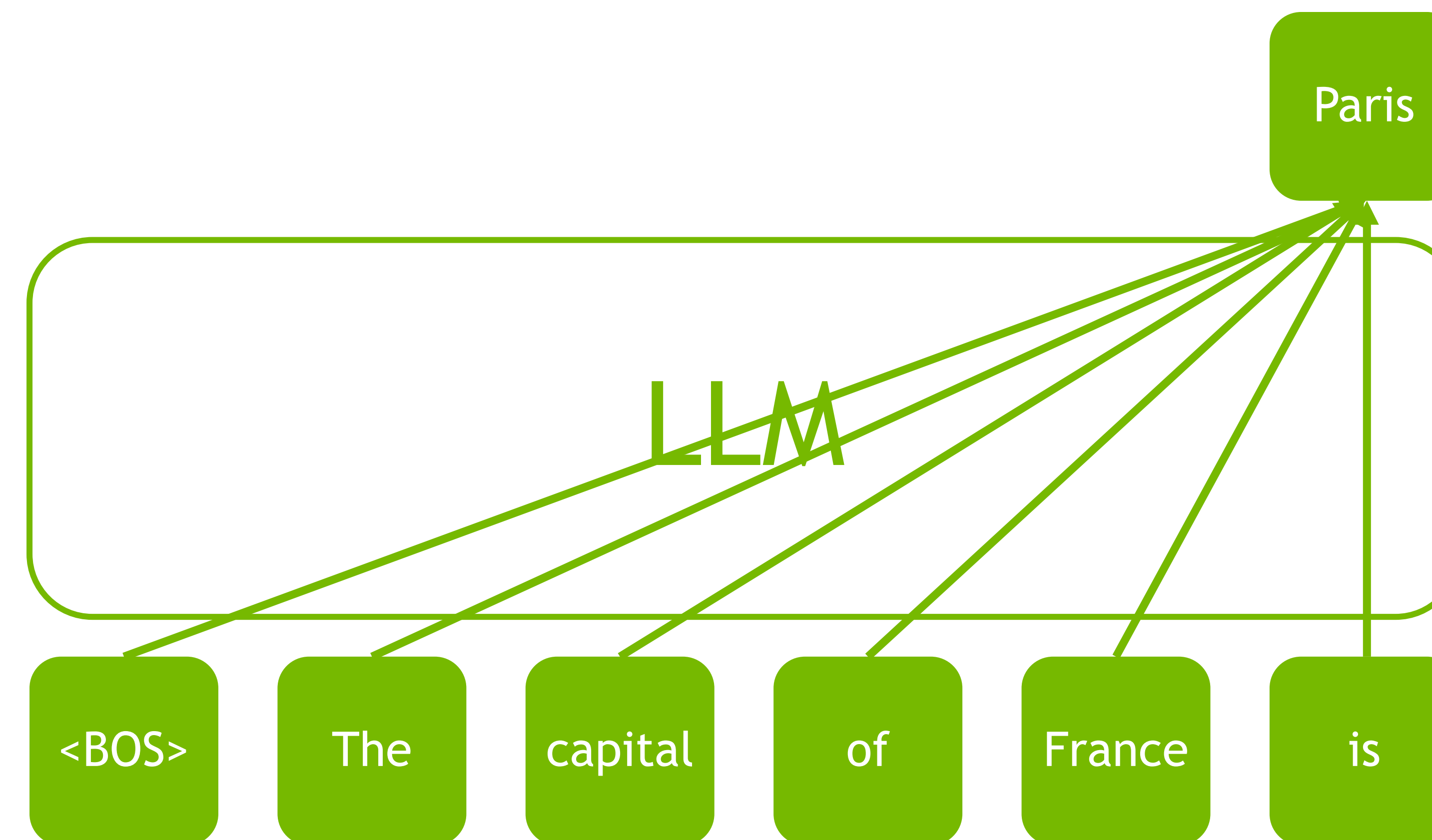
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Supervised Fine-tuning

- Learning from **high-quality labeled** data
- **Supervised Learning** through **next unit prediction**
- Teaches the model how to behave
 - **Follow instructions**
 - Give helpful answers
 - Use the right tone and format

LLM

What

is

the

capital

of

France

?

<EOS>

The

capital

of

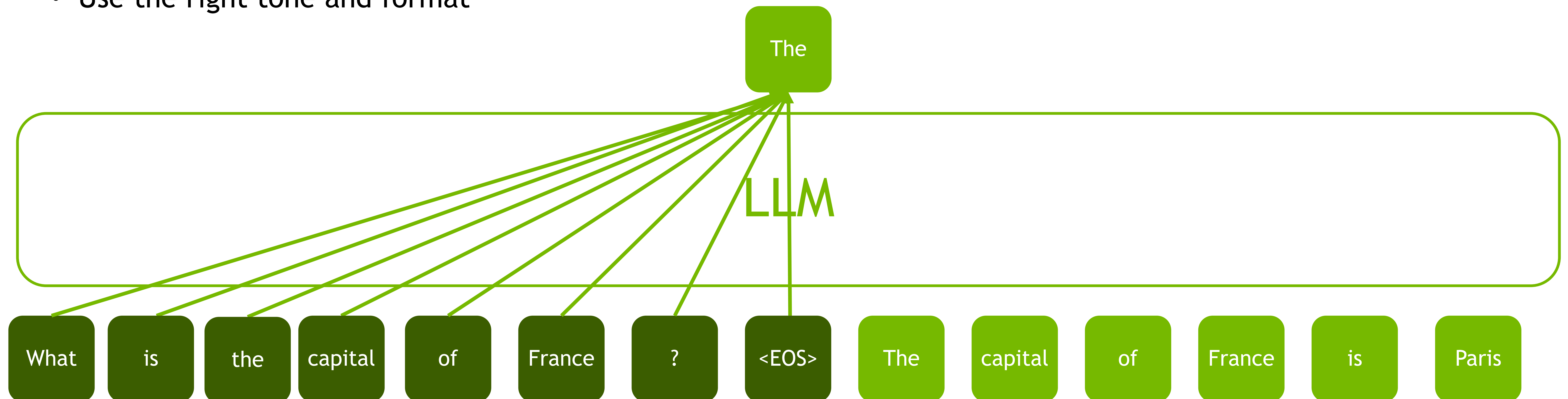
France

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Paris

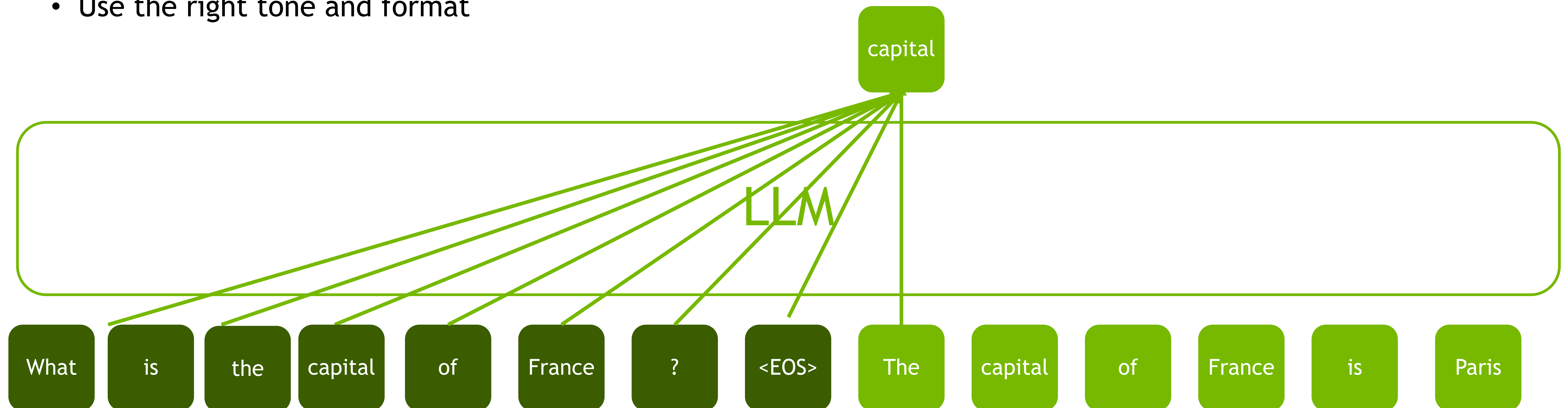
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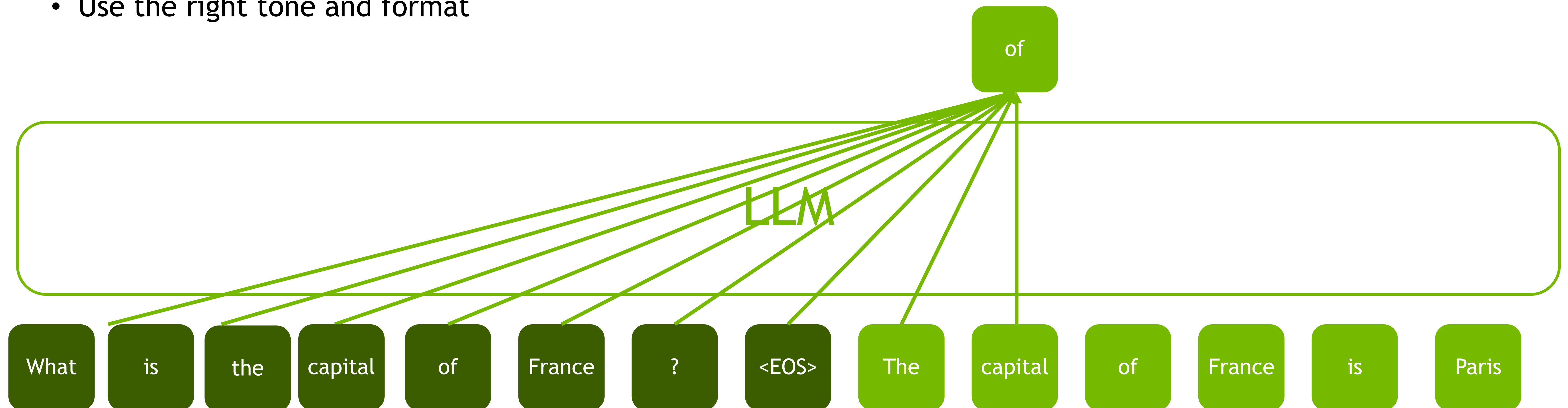
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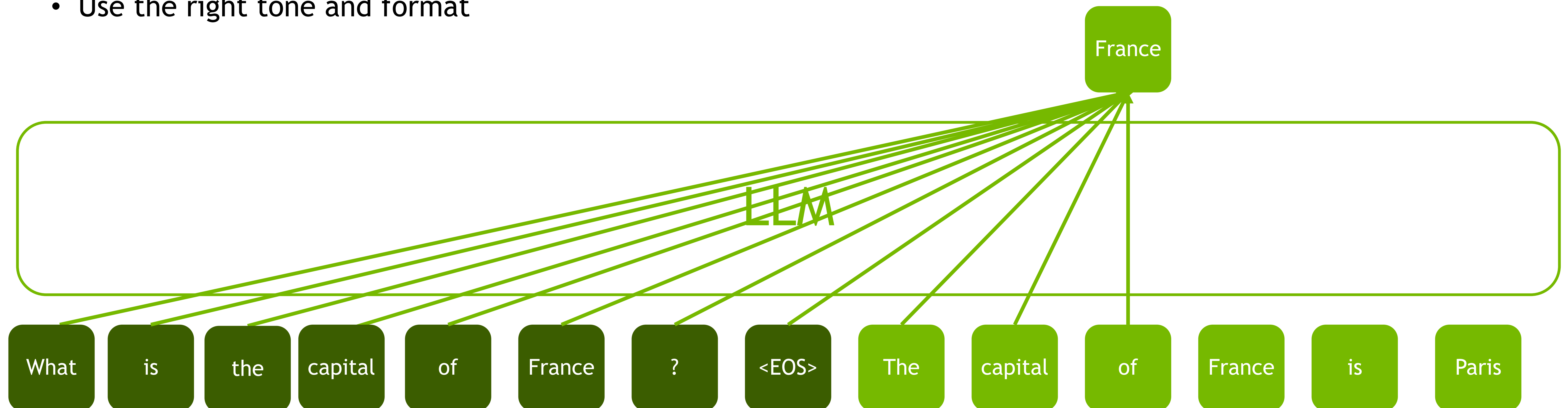
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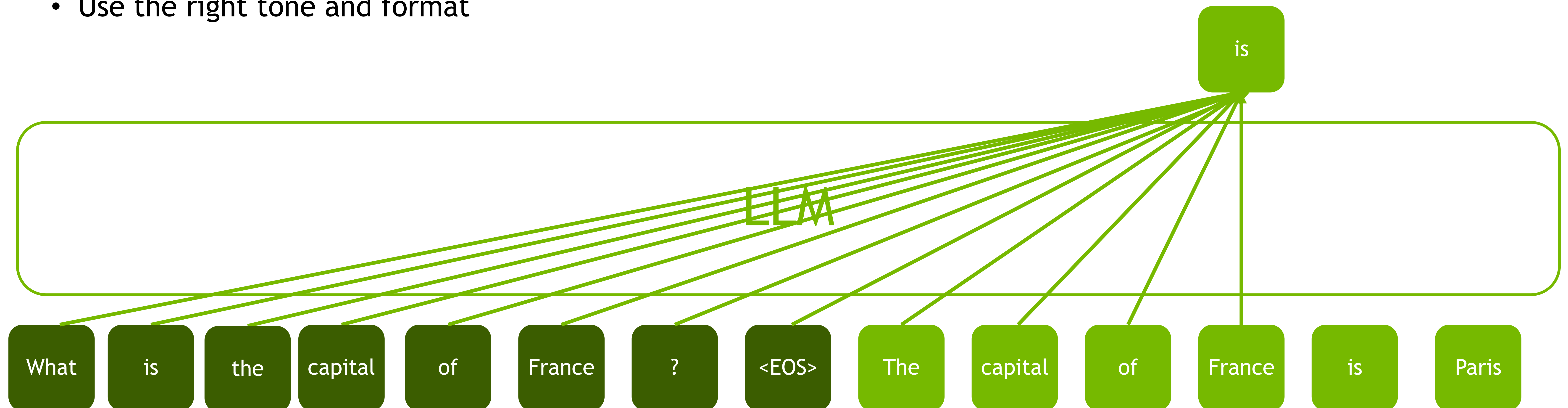
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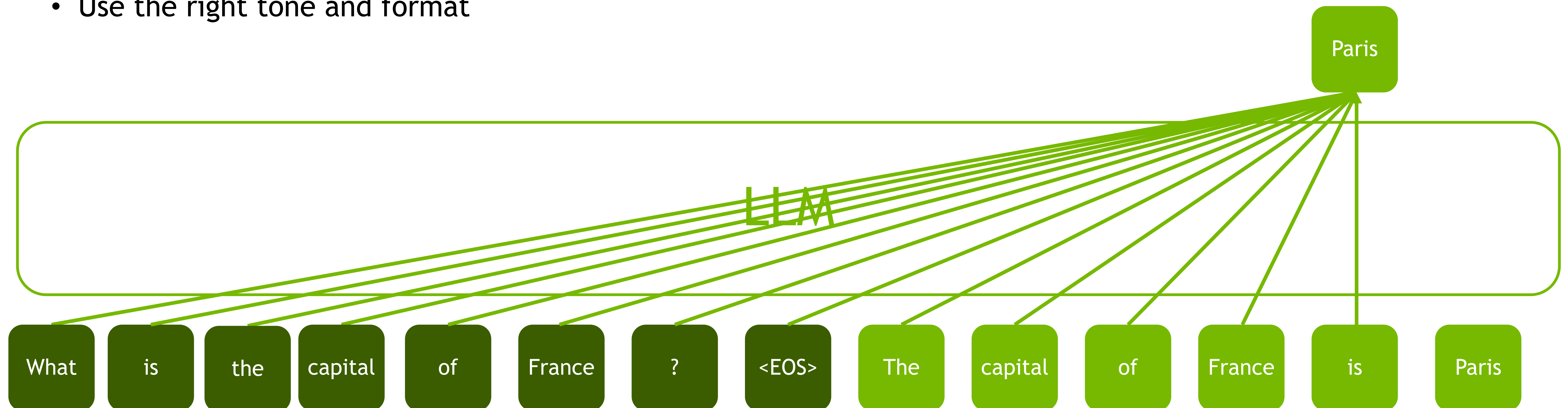
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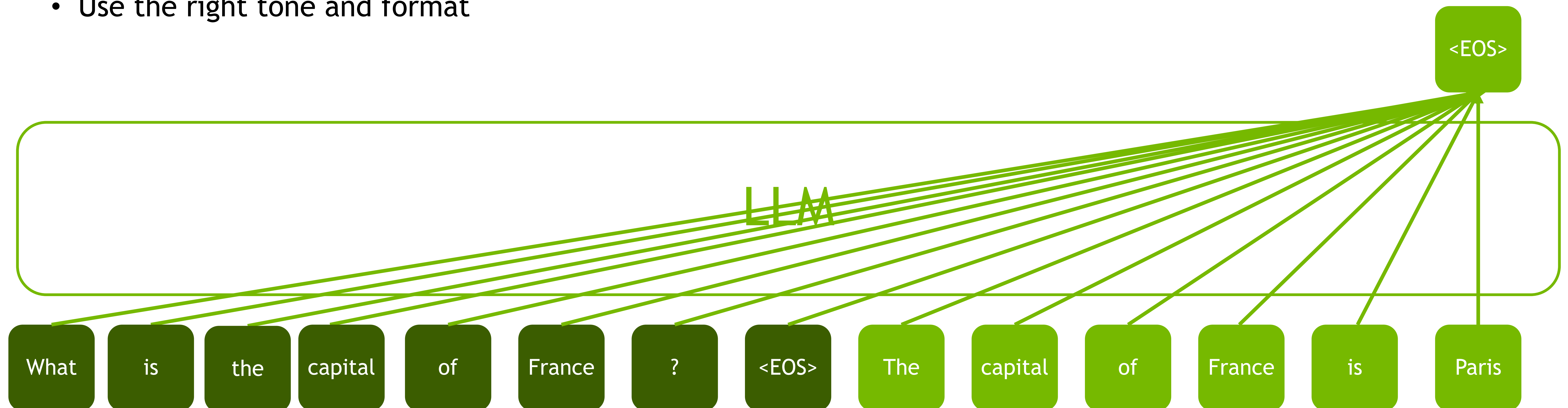
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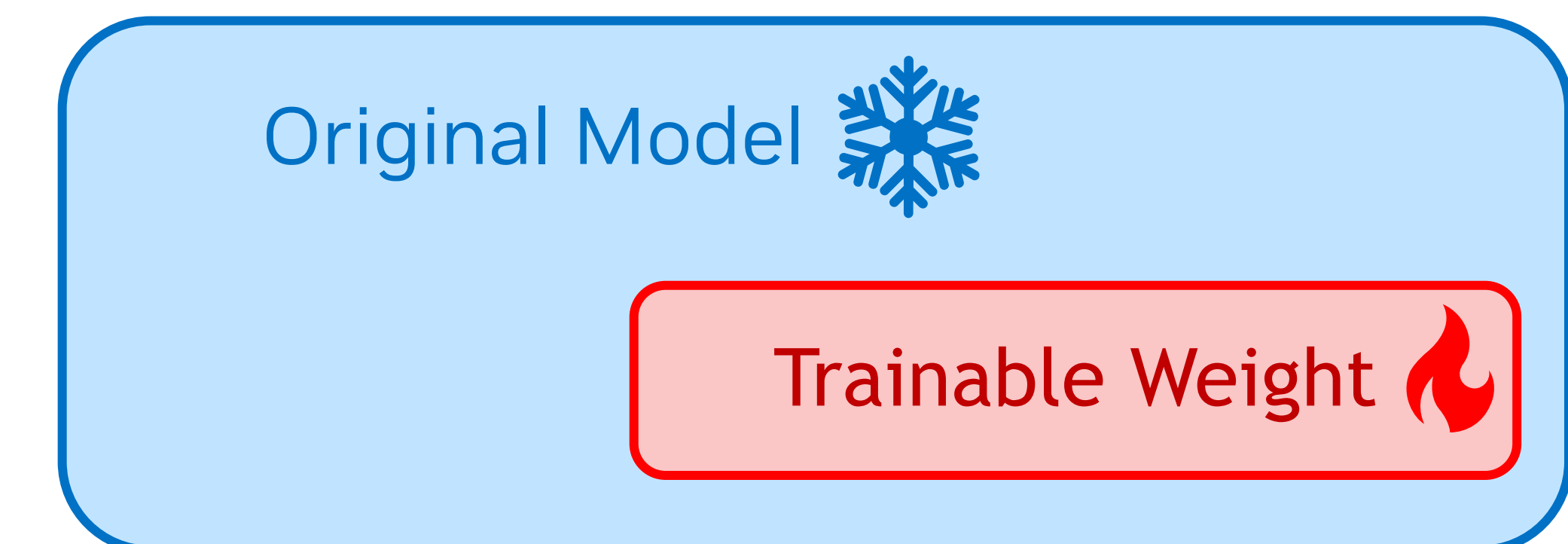
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Supervised Fine-tuning (PEFT)

- Full fine-tuning requires a lot of computing resources and data to make the model converge
- **Parameter Efficient Fine-tuning (PEFT) freeze** most of the model parameters, and add a **small subset of parameters**
- When to use PEFT:
 - **Limited computing resources**
 - **Limited labeled data**
 - **Prototyping**





NeMo AutoModel

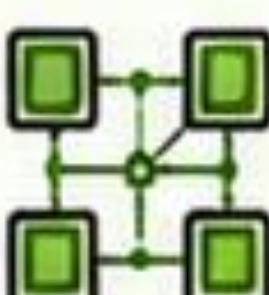
NeMo AutoModel Overview

An end-to-end framework for training and fine-tuning large models at any scale



What is it?

NVIDIA framework for training and fine-tuning LLMs and VLMs.



Built on PyTorch (distributed-first)

Same code runs from 1 GPU to 1000+ GPUs.



Supports full ML workflow

- Pre-training
- Supervised Fine-Tuning (SFT)
- Parameter-Efficient Fine-Tuning (PEFT: LoRA, QLoRA)



Easy to use & flexible

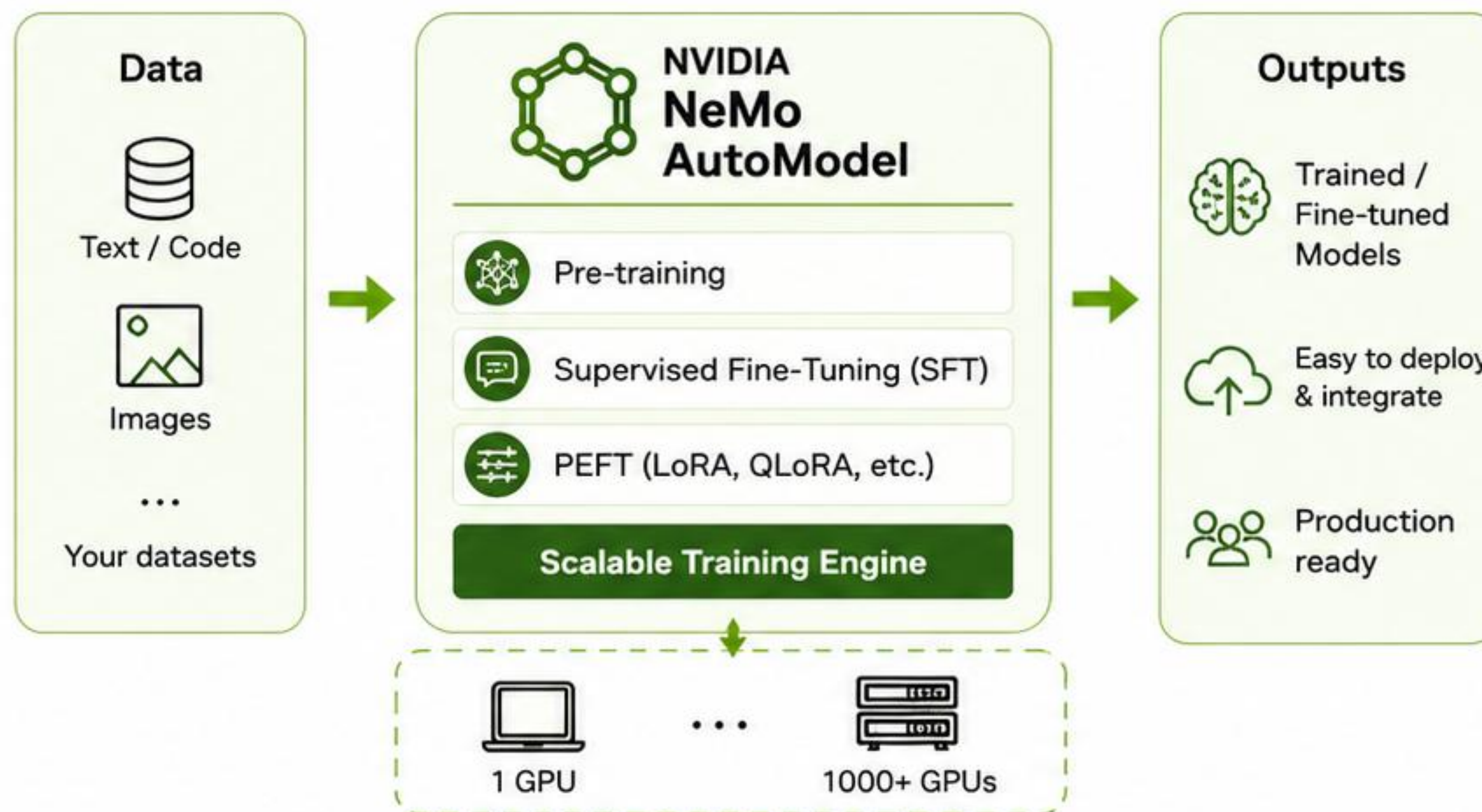
- YAML-based configs ("recipes")
- Works with Hugging Face models out-of-the-box



Scales efficiently

- Distributed training (FSDP, tensor & pipeline parallelism)
- Optimized for large models (multi-node, multi-GPU)

From data to optimized models



Why NeMo AutoModel?



Faster time to train



Cost-efficient training at scale



Production-ready workflows



Seamless with NVIDIA ecosystem

NeMo AutoModel makes training and fine-tuning large models easier, faster, and more scalable.